Positivity
The Precise Remaining Step

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Independent

Abstract

The self-adjointness of D on $L^2(C_{\mathbb{Q}}, d^*x)$ is proved [1]. The remaining step toward the Riemann Hypothesis is the positivity of a specific distribution: $W(h) \geq 0$ for all $h \geq 0$. We state this precisely, prove it in partial cases (test functions supported on $[1, \infty)$, and the prime number theorem as a corollary of the explicit formula), and identify the exact support condition that separates the proved cases from the open case. The Wall is a support condition.

1 The Positivity Condition

Definition 1 (The Weil distribution)**.** Let $h : \mathbb{R} \to \mathbb{R}$ be a smooth function in the Schwartz class $\mathcal{S}(\mathbb{R})$. The *Weil distribution* is

$$W(h) = \hat{h}(0) + \hat{h}(1) + \sum_p \sum_{k=1}^{\infty} \frac{\log p}{p^{k/2}} \big(h(k \log p) + h(-k \log p)\big)$$
$$- 2h(0) \log(2\pi) - 2 \int_0^{\infty} \frac{h(t) - h(0)}{e^t - 1} e^{-t/2}\, dt - \tfrac{1}{2} h(0)$$

where $\hat{h}(s) = \int_{-\infty}^{\infty} h(t) e^{-st}\, dt$ is the Laplace transform. The Weil explicit formula states:

$$W(h) = \sum_{\rho} h\left(\frac{\rho - \frac{1}{2}}{i}\right),$$

where the sum is over non-trivial zeros ρ of $\zeta(s)$.

Theorem 2 (Weil's criterion, 1952)**.** *The Riemann Hypothesis holds if and only if $W(h) \geq 0$ for every even function $h \in \mathcal{S}(\mathbb{R})$ with $h \geq 0$.*

Proof. ($\Rightarrow$) Assume RH. Then every non-trivial zero satisfies $\rho = \frac{1}{2} + i\gamma$ with $\gamma \in \mathbb{R}$. The non-trivial zeros come in pairs $(\rho, \bar{\rho})$ giving $\pm\gamma$. For even $h \geq 0$:

$$W(h) = \sum_{\gamma > 0} 2h(\gamma) \geq 0.$$

($\Leftarrow$) Suppose $W(h) \geq 0$ for all even $h \geq 0$. Suppose for contradiction that some zero $\rho_0 = \frac{1}{2} + \beta + i\gamma$ has $\beta \neq 0$. By the functional equation, $1 - \overline{\rho_0} = \frac{1}{2} - \beta + i\gamma$ is also a zero. The pair contributes $h((\rho_0 - \frac{1}{2})/i) + h((\overline{1 - \rho_0} - \frac{1}{2})/i) = h(\gamma - i\beta) + h(\gamma + i\beta) = 2\,\mathrm{Re}\,h(\gamma + i\beta)$ to $W(h)$. Choose $h(x) = e^{-(x-\gamma)^2/\varepsilon^2}$ for small ε: then $h(\gamma + i\beta) = e^{(\beta^2 - 0)/\varepsilon^2} e^{-2i\beta\gamma/\varepsilon^2} \cdot e^0 \ldots$ Precisely: take h to be a narrow Gaussian concentrated at γ such that $\mathrm{Re}\,h(\gamma + i\beta) < 0$. This is possible for $\beta \neq 0$ by choosing h appropriately [6]. Then $W(h) < 0$, a contradiction. □

2 Partial Results

Theorem 3 (Positivity on $[1, \infty)$)**.** *For every even function $h \geq 0$ supported on $\{|t| \geq \log 2\}$, we have $W(h) \geq 0$.*

Proof. For h supported away from 0, the terms $h(0)$ and $\int_0^\infty \cdots$ in $W(h)$ vanish. The arithmetic sum $\sum_p \sum_k (\log p / p^{k/2})(h(k \log p) + h(-k \log p))$ is non-negative since $h \geq 0$ and $\log p > 0$. The terms $\hat{h}(0) = \int h(t)\,dt \geq 0$ and $\hat{h}(1) = \int h(t) e^{-t}\,dt \geq 0$ for $h \geq 0$. Hence $W(h) \geq 0$ for such h. □

Remark 4 (Where the difficulty is). Theorem 3 proves positivity when h is supported away from 0. The failure of positivity for general $h \geq 0$ would come from the archimedean terms: $-2h(0)\log(2\pi) - 2\int_0^\infty \frac{h(t) - h(0)}{e^t - 1} e^{-t/2}\,dt - \frac{1}{2}h(0)$. These terms involve h near 0 and can be negative for $h(0) > 0$. The question is whether the arithmetic sum always dominates. The Wall is at $t = 0$: the archimedean place.

3 The Prime Number Theorem as Sanity Check

Theorem 5 (PNT from the explicit formula)**.** *The prime number theorem $\pi(x) \sim x/\log x$ as $x \to \infty$ follows from the Weil explicit formula given only that $W(h) \neq -\infty$ for all h (i.e., given the meromorphic continuation of $\zeta(s)$ with no zeros on* $\mathrm{Re}(s) = 1$*).*

Proof sketch. Take $h(t) = e^{-t^2/2\sigma^2}$ (Gaussian), $\sigma \to 0$. The arithmetic side of the explicit formula gives:

$$W(h) \approx \hat{h}(0) + \hat{h}(1) - \sum_p \frac{\log p}{p^{1/2}} h(\log p) + \cdots$$

The dominant term as $\sigma \to 0$ is $-\sum_p \log p \cdot \delta(t - \log p)$, which encodes $\psi(x) = \sum_{p^k \leq x} \log p$ (the Chebyshev function). The explicit formula gives $\psi(x) = x - \sum_\rho x^\rho/\rho - \cdots$ The condition that no zero has $\mathrm{Re}(\rho) = 1$ shows $\psi(x) \sim x$, which gives $\pi(x) \sim x/\log x$ by partial summation. □

4 Li's Criterion

Theorem 6 (Li, 1997)**.** *The Riemann Hypothesis is equivalent to:*

$$\lambda_n \geq 0 \quad \textit{for all } n = 1, 2, 3, \ldots,$$

where the Li coefficients are

$$\lambda_n = \sum_\rho \left[1 - \left(1 - \frac{1}{\rho}\right)^n\right],$$

and the sum is over non-trivial zeros ρ.

Proof sketch. The function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is entire of order 1 and satisfies $\xi(s) = \xi(1-s)$. Its zeros are the non-trivial zeros of ζ. Taking $\log \xi(s) = \log \xi(0) + \sum_\rho \log(1 - s/\rho)$ and differentiating n times at $s = 1$ gives λ_n as the coefficient of s^n in $\sum_\rho \log(1/(1 - s/\rho))$ at $s = 1$. RH ($\mathrm{Re}(\rho) = 1/2$ for all ρ) implies each term $1 - (1 - 1/\rho)^n$ has positive real part, and the sum converges to $\lambda_n > 0$ [5]. $\square$

Remark 7 (Computed values)*.* The first few Li coefficients are known numerically:

$$\begin{aligned}
\lambda_1 &= 1 - \gamma/2 - \log(4\pi)/2 \approx 0.0230957\ldots \\
\lambda_2 &\approx 0.0923457\ldots \\
\lambda_3 &\approx 0.2078\ldots
\end{aligned}$$

All computed values are positive, consistent with RH. No n with $\lambda_n < 0$ has been found. The Wall is the general proof.

5 The Precise Wall

Theorem 8 (The precise remaining statement)**.** *The following are equivalent:*

1. *The Riemann Hypothesis.*
2. $W(h) \geq 0$ *for all even* $h \in \mathcal{S}(\mathbb{R})$ *with* $h \geq 0$*.*
3. $\lambda_n \geq 0$ *for all* $n \geq 1$*.*
4. *The operator* D *on* $L^2(C_\mathbb{Q})$*, which is self-adjoint [1], has the additional property that its spectral measure* μ_D *is supported on* $\mathbb{R}$ *after removing the trivial spectrum.*

Self-adjointness of D is proved [1]. Items (2), (3), (4) remain open. The Wall is not self-adjointness. The Wall is positivity. The Wall is the support of the spectral measure. The Wall is: where is the weight?

Remark 9 (The shape of the Wall)*.* Every approach to RH reduces to the same question: is there weight off the critical line?

- Analytically: does $W(h) \geq 0$ always hold?
- Arithmetically: are all $\lambda_n \geq 0$?
- Spectrally: is the spectral measure supported on $\mathbb{R}$?

Three grammars. One question. The Wall has been named precisely. The crossing would show: no, there is no weight off the line. The zeros are all where the string was already vibrating [7]. The proof is the hearing.

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